

Name: _____ Date: _____

Period: _____

Primary Objective

By the end of this lesson, I will be able to...

1. Rewrite a polynomial or rational expression in **factored** form or **standard** form and explain what each form reveals. (LO 1.11.A)
2. Perform polynomial **long division** to write $f(x) = g(x)q(x) + r(x)$ and use the result to find **slant** asymptotes. (LO 1.11.B)
3. Apply the **binomial** theorem and Pascal's Triangle to expand $(a + b)^n$. (LO 1.11.C)

Vocabulary & Key Concepts**factored form:**

A polynomial or rational expression written as a product of linear and/or irreducible quadratic factors...... **standard form:**

A polynomial written with terms in descending order of degree; reveals degree, leading coefficient, and end behavior...... **polynomial long division:**

An algorithm for dividing polynomial f by polynomial g to find quotient q and remainder r such that $f = gq + r$ **quotient:**

The polynomial $q(x)$ resulting from dividing $f(x)$ by $g(x)$ in long division...... **remainder:**

The polynomial $r(x)$ left over after long division, with $\deg(r) < \deg(g)$ **slant asymptote:**

A linear asymptote $y = q(x)$ that occurs when the degree of the numerator is exactly one more than the degree of the denominator...... **binomial theorem:**

The formula $(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$, which uses Pascal's Triangle coefficients...... **Pascal's Triangle:**

A triangular array of numbers where each entry is the sum of the two entries above it; row n gives the coefficients for $(a + b)^n$.

Hook

Three rational functions are written on the board:

$$f(x) = \frac{2x^2 - 8}{x^2 + x - 6} \quad g(x) = \frac{2(x-2)(x+2)}{(x-2)(x+3)} \quad h(x) = \frac{2(x+2)}{x+3}, \quad x \neq 2$$

These are the *same function* in three different faces. **Factored** form reveals zeros and holes. **Standard** form reveals end behavior. The simplified form reveals the **domain restriction** of the restricted domain.

LO 1.11.A — Factored Form vs. Standard Form

	Factored Form	Standard Form
What it reveals	zeros, holes, asymptotes, domain zeros, holes, asymptotes	end behavior, degree, leading coefficient end behavior, degree, leading coefficient
Example (poly)	$(x-1)(x+3)(x-2)$	$x^3 - 7x + 6$ (expand)
Example (rational)	$\frac{(x-2)(x+2)}{(x-2)(x+3)}$	$\frac{x^2 - 4}{x^2 + x - 6}$

Key idea (EK 1.11.A.1): Factored form reveals **zeros, asymptotes, holes**, and domain restrictions.

Key idea (EK 1.11.A.2): Standard form reveals **end behavior and degree**.

LO 1.11.B — Polynomial Long Division

Division Algorithm: If f is divided by g (with $\deg g \geq 1$), then

$$f(x) = g(x) \cdot q(x) + r(x), \quad \text{where } \deg(r) < \deg(g).$$

Steps:

1. Write numerator and denominator in **descending** order (fill gaps with 0 coefficients).
2. Divide the **leading** term of the dividend by the leading term of the divisor. Write this as the first term of $q(x)$.
3. Multiply that term by the **entire divisor** and subtract from the dividend.
4. Bring down the next term and repeat until $\deg(\text{remainder}) < \deg(\text{divisor})$.

Example: Divide $f(x) = 2x^2 + 3x - 5$ by $g(x) = x - 1$.

$$\begin{array}{r}
 2x + 5 \\
 x - 1 \overline{) 2x^2 + 3x - 5} \\
 \underline{-(2x^2 - 2x)} \\
 5x - 5 \\
 \underline{-(5x - 5)} \\
 0
 \end{array}$$

Result: $q(x) = 2x + 5$, $r(x) = 0$, so $f(x) = (x - 1)(2x + 5) + 0$.

Slant Asymptote (EK 1.11.B.2): When $\deg(f) = \deg(g) + 1$, the slant asymptote of $\frac{f(x)}{g(x)}$ is $y = q(x)$ (the **quotient polynomial** after long division).

LO 1.11.C — Binomial Theorem and Pascal's Triangle

Pascal's Triangle (fill in the missing entries):

			1			
			1		1	
		1		2		1
	1		3		3	
1		4		6		4
	1		3		3	
		1		1		

Rule: Each entry equals the **sum** of the two entries above it.

Binomial Theorem (EK 1.11.C.1):

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$

The coefficients $\binom{n}{k}$ match row n of Pascal's Triangle.

Example: Expand $(x + 2)^3$.

Using row 3 of Pascal's Triangle (1, 3, 3, 1):

$$(x + 2)^3 = 1 \cdot x^3 + 3 \cdot x^2(2) + 3 \cdot x(4) + 1 \cdot (8) = x^3 + 6x^2 + 12x + 8$$